



Development of 8-branch Er: fiber frequency comb for Sr and Yb optical lattice clocks

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Abstract: We demonstrate an 8-branch Er: fiber frequency comb with seven application ports, which can be individually optimized for applications with different wavelengths. The beat between the comb and a cw laser has a signal-to-noise ratio exceeding 30 dB at a resolution bandwidth of 300 kHz. The 8-branch frequency comb is used to perform frequency locking for four repumping and lattice lasers, and the frequency measurement of two clock lasers of strontium and ytterbium optical lattice clocks. We have achieved reliable optical lattice clock operation, thanks to the stable frequency locking and measurement obtained by using the 8-branch frequency comb. The developed frequency comb is a powerful experimental tool for various applications, including not only optical lattice clocks, but also research on quantum optics that use many frequency-stabilized lasers.

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1. Introduction

Optical frequency metrology [1] is of great interest for a wide range of applications, including the fundamental science and technologies that support precision measurement and broadband communication networks. The invention of optical frequency combs [2,3] has revolutionized the field of optical frequency measurement and further stimulated research on optical frequency standards [4]. To date, optical frequency standards based on single ions (single ion clocks) and neutral atoms trapped in optical lattices (optical lattice clocks) have achieved unprecedented fractional uncertainties at the 10^{-18} level [5–9]. Optical frequency standards are considered to be next-generation atomic clocks for a redefinition of the SI second [10,11] and also a powerful tool for relativistic geodesy [12–14]. The research and development of optical clocks and combs is mutually supportive in terms of making further improvements and finding new applications.

Recently several institutes have conducted direct comparisons of optical clocks and reported the frequency ratios of the clock transitions. These results help to confirm clock consistency at a level much better than that of Cs clocks [15] and to test the fundamental physics [16–18]. Of the several frequency ratios obtained using different atomic species [14,16,19–25], the ratio of Sr and Yb optical lattice clocks [14,20–22] is the only one that has been both investigated at different institutes and shown an uncertainty at the 10^{-17} level. Moreover, the dual-mode operation of Sr and Yb optical lattice clocks in a single chamber has recently been reported [26]. To operate Sr and Yb clocks simultaneously, a number of cooling, repumping and lattice lasers need to be frequency stabilized. Most of these lasers have often been stabilized using their respective optical cavities. However, the frequency of the optical cavity usually drifts depending on the environmental temperature, and the laser frequency is sometimes unlocked due to environmental vibrations affecting the cavity.

Instead, an optical frequency comb can be used to lock multiple lasers with much better frequency stability and also system reliability.

Optical frequency combs based on Er: fiber mode-locked lasers have advantages in that they are more compact, robust and cost efficient than frequency combs based on other mode-locked lasers. Specifically, an Er: fiber laser using a nonlinear polarization rotation (NPR) mode-lock scheme [27,28] can be assembled with standard telecom parts and fiber splicers. Therefore, it is also easy to develop a multi-branch Er: fiber frequency comb [28] with each branch optimized to meet the requirements of each application. Fiber combs with up to 4 branches have already been reported. It should be noted that since one of the branches is usually used to detect and stabilize the carrier envelope offset frequency (f_{CEO}) [3], only the remaining 3 branches can be used as application ports. When Sr and Yb optical clocks are operated, more than three different lasers need to be frequency stabilized. Therefore, it is useful to reveal the technical details of a fiber comb with more than 4 branches. It is also interesting to test whether or not such a multi-branch comb system is sufficiently reliable for the operation of optical clocks.

In this study, we demonstrate a frequency comb with 8 branches using 3 stages consisting of 1:1 fiber couplers, where 7 of the branches can be used as application ports. Each branch is individually optimized to lock the cooling (798 nm), repumping (1389 nm and 679 nm) and lattice (759 nm and 813 nm) lasers, and to measure the clock (578 nm and 698 nm) lasers of Sr and Yb optical lattice clocks [see Fig. 1]. The beat between the comb and a cw laser has a signal-to-noise ratio (S/N) exceeding 30 dB at a resolution bandwidth of 300 kHz. We perform frequency locking for 4 repumping and lattice lasers, and measure the frequency of 2 clock lasers of Sr and Yb optical lattice clocks to check the overall reliability of our developed 8-branch frequency comb.

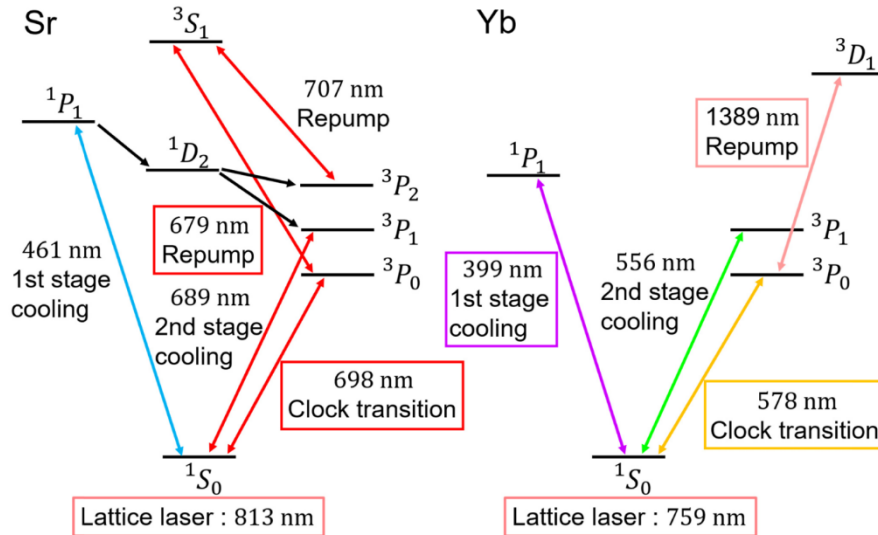


Fig. 1. Energy diagram for Sr and Yb optical lattice clocks. Wavelengths are indicated for the relevant cooling, repumping and clock transitions, and also lattice lasers. For those transitions highlighted with squared boxes, the light sources are locked or measured using an 8-branch frequency comb.

2. Experimental setup

Figure 2 shows a schematic diagram and a photo of an 8-branch Er: fiber frequency comb. The oscillator is based on an NPR mode-locked laser with the cavity design described in [28]. The length of the Er: fiber in the oscillator is 90 cm. The peak absorption of the Er: fiber at 1530 nm is approximately 40 dB/m. The repetition frequency (f_{rep}) of the mode-locked laser is

approximately 50.25 MHz. A fiber coupler (99:1) followed by an isolator is introduced after the laser cavity for f_{rep} detection. To transfer the frequency stability of the coordinate universal time at National Metrology Institute of Japan (UTC(NMIJ)) to f_{rep} without degradation [29], we mix the 20th harmonic frequency (~ 1005 MHz) of f_{rep} with a 1 GHz reference signal generated by multiplying the 10 MHz reference signal by a factor of 100. The intermediate output from the mixer (~ 5 MHz) is then phase-locked to a reference frequency generated by a synthesizer (Keysight 33500B) by controlling the cavity length of the oscillator using a piezoelectric transducer (fast servo) and a Peltier element (slow servo) as shown in Fig. 2. The synthesizer is also referenced to UTC(NMIJ).

The oscillator output power of 2.9 mW is equally distributed to 8 branches with 3 stages of fiber couplers. Each branch has a polarization controller containing both half and quarter waveplates, and an Er-doped fiber amplifier (EDFA) pumped by a 980-nm laser diode (LD). The dispersion is compensated for in each branch so that the output power is amplified and the spectrum is broadened through the EDFA [30]. The output power is approximately 70 mW at a pump LD injection current of 700 mA. Before coupling the output from the fiber to free space, an isolator is inserted to prevent any possible back-reflected light.

One of the branches is used to detect the f_{CEO} of the frequency comb using an f - $2f$ interferometer [3]. A highly nonlinear fiber (HNLF) following the EDFA broadens the spectrum to one octave for f_{CEO} detection. f_{CEO} is phase locked to another synthesizer (Keysight 33500B) referenced to UTC(NMIJ). The error signal is fed back to the injection current of the 1480-nm pump LD for the oscillator.

The other 7 branches are application branches, which are used for experiments on Sr and Yb optical lattice clocks. For example, one of the branches (shown in Fig. 2) is used to stabilize the frequency of the lattice laser of the Sr clock at a “magic” wavelength of 813 nm, where the clock transition frequency is not disturbed by the lattice laser. The amplified comb light was frequency-doubled from 1626 nm to 813 nm using a bulk periodically poled lithium niobate (PPLN) crystal. A beat frequency (f_{beat}) is observed between the frequency-doubled comb mode and a cw Ti:sapphire laser (M-Squared Lasers; SolsTiS-SRX). An optical filter or a grating is used to remove the comb components that do not contribute to the beat measurement, and hence increase the S/N of the heterodyne beat. The observed f_{beat} is used to frequency stabilize the Ti:sapphire laser to the comb based on the electronic delay-line method [31,32]. Three more branches are used for laser frequency stabilization using the same delay-line method. One of them is used to frequency stabilize the lattice laser of the Yb lattice clock at a “magic” wavelength of 759 nm. Two branches are used to stabilize the repumping lasers for exciting atoms from the metastable state $^3\text{P}_0$ (679 nm for Sr and 1389 nm for Yb). One branch will be used to stabilize a laser at 798 nm, which is the fundamental light used to generate 399 nm-light for the 1st stage cooling of Yb [33].

The remaining two branches are used for the absolute frequency measurement of the clock transition $^1\text{S}_0$ - $^3\text{P}_0$ of Sr (698 nm) and Yb (578 nm). All the beat signals (f_{CEO} and f_{beat}) are monitored using a spectrum analyzer and measured using a dead-time free frequency counter (Pendulum CNT-90).

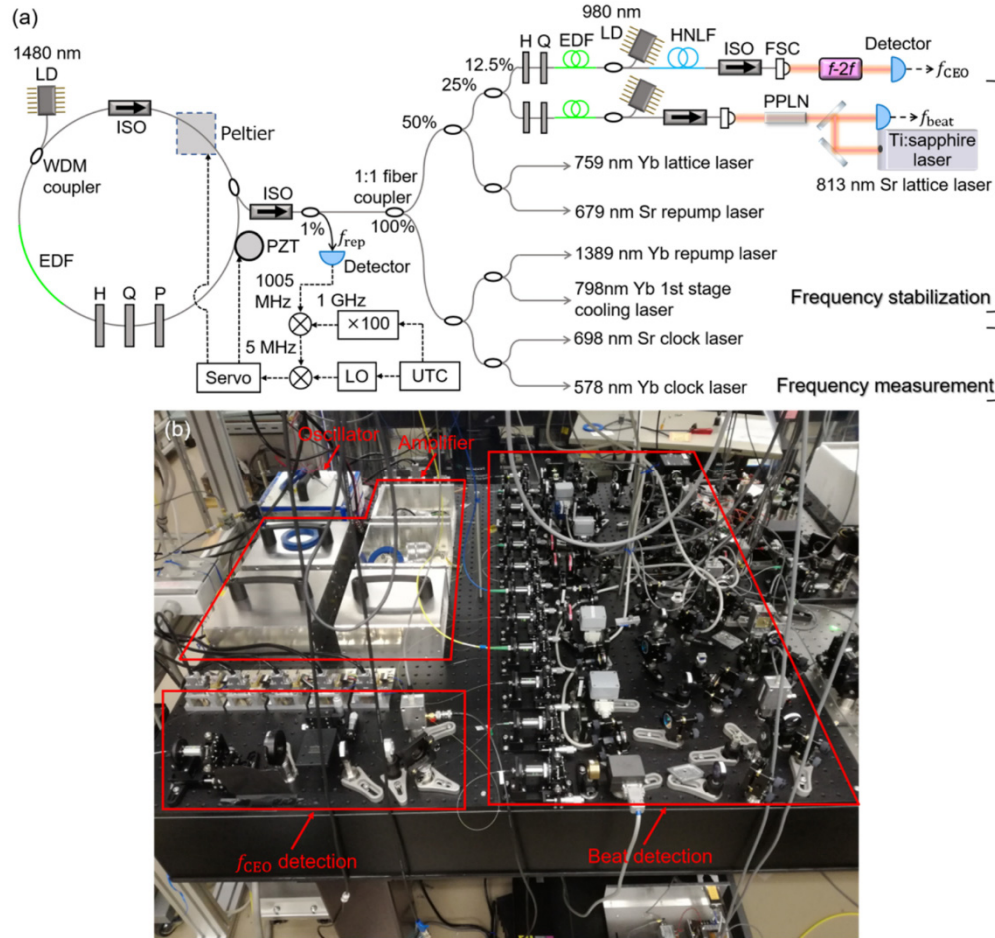


Fig. 2. (a) Schematic diagram of an 8-branch Er:fiber frequency comb and its application for Sr and Yb optical lattice clocks. LD: laser diode, EDF: Er-doped fiber, H: half-wavelength plate, Q: quarter-wavelength plate, P: polarizer, PZT: piezoelectric transducer, ISO: isolator, FSC: free-space coupler, HNLF: highly nonlinear fiber, f -2 f : f -2 f interferometer, PPLN: periodically poled lithium niobate. (b) Photograph of 8-branch fiber comb.

3. Experimental results

3.1. Phase-locking of f_{CEO} and f_{rep}

The inset in Fig. 3 shows the f_{CEO} and $(f_{\text{rep}} - f_{\text{CEO}})$ beat signals observed using a spectral analyzer. The S/N of the beat signals was 32 dB at a resolution bandwidth of 300 kHz, which is sufficiently high for phase-locking without any cycle slips. In the present experiment, the beat signal $f_{\text{rep}} - f_{\text{CEO}}$ around 30 MHz was frequency-divided to 3 MHz, and then phase-locked to a synthesizer referenced to UTC(NMIJ). The frequency division by a factor of 10 was introduced to extend the capture range in the analog phase locking using a mixer. Consequently, this increases the reliability of the locking. Figure 3 shows the phase-locked f_{CEO} recorded using a frequency counter over 1 million seconds. The counter gate time is 1 s. No unexpected instantaneous or long-term frequency variation was observed in the recorded f_{CEO} . We succeeded in phase-locking f_{CEO} for more than 2 weeks until we switched off the system.

We note that the configuration of the f_{CEO} branch is exactly the same as that described in [28], except for the isolator installed before the free-space coupler. We found that the output

of the EDFA was significantly low (~ 20 mW) without the isolator. The output of the EDFA increased to 70 mW, which is sufficient for spectral broadening using HNLF, when we inserted the isolator or changed the fiber connector in the coupler from a physical contact (PC) to an angled physical contact (APC) connector. When the input power of the EDFA is low the back reflected light may compete with the input light and cause a decrease in the amplification in the EDFA.

We have also succeeded in phase-locking f_{rep} for more than 2 weeks. Since the measurement of the in-loop f_{rep} signal only reflects the counter noise, the results of the f_{rep} phase-locking are discussed in Section 3.4 by comparing the optical lattice clocks and UTC(NMIJ).

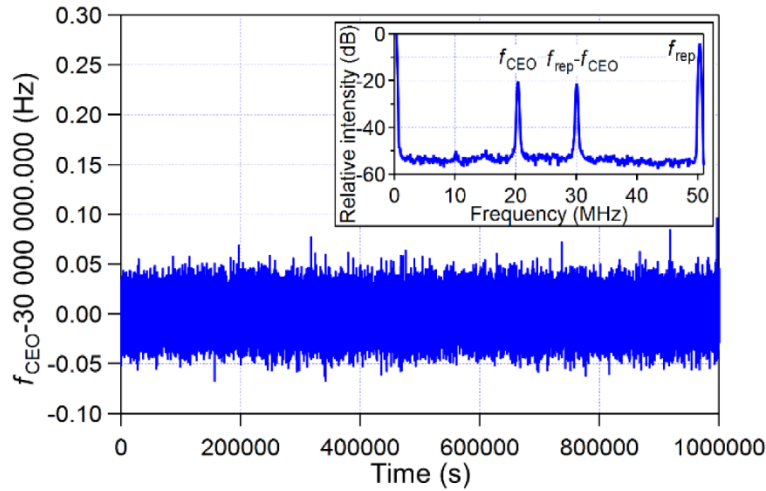


Fig. 3. Recorded frequency value of the phase-locked f_{CEO} as a function of time. The inset shows the observed f_{CEO} signals at a resolution bandwidth of 300 kHz. f_{rep} is the repetition rate of the fiber comb. ($f_{\text{rep}} - f_{\text{beat}}$) is the beat frequency between the laser and the second-nearest comb mode.

3.2. Broadened frequency comb spectra and beat detection

Figure 4 shows the spectra of the 7 application branches of the comb observed with an optical spectrum analyzer. To avoid any overlap, the spectra are displayed with offsets from bottom to top in the figure as the spectrum bandwidth increases. With an EDFA, the comb spectrum is usually broadened to cover a wavelength range from 1.5 to 1.6 μm (the bottom three spectra). With a combination of EDFA + HNLF, the comb spectrum is able to cover a wider wavelength range of up to one octave from 1.0 to 2.0 μm (the top four spectra). To access shorter wavelengths of 500 nm to 1.0 μm , a bulk PPLN crystal is useful for frequency doubling the comb components. The phase-matching condition allows the frequency-doubled comb modes to be generated only at the targeted wavelength. In the present experiment, the output of each branch, except for the branch for the laser at 1389 nm, is frequency-doubled with a PPLN to access the targeted wavelength.

Each spectrum of the 7 branches is tailored by individually controlling the light polarization and the selection of the HNLF, where it applies, to obtain the maximum comb light output at a desired wavelength. We use HNLFs manufactured by Sumitomo Electric Industries. A dispersion flattened HNLF is usually used to generate flat and smooth spectra, but with a limited wavelength range. In the present experiment, a dispersion flattened HNLF is used in the 698-nm and 679-nm branches. In the 1389-nm branch, we used a standard HNLF with a dispersion slope of 0.020 ps/nm²-km and a zero dispersion wavelength of 1565 nm. Both HNLFs worked well in this wavelength region. To generate a much shorter wavelength at 1156 nm in the 578-nm branch, we used a standard HNLF with a larger

dispersion slope ($0.025 \text{ ps/nm}^2\text{-km}$) and a shorter zero dispersion wavelength (1545 nm). We note here that, to generate a broader spectrum in the f_{CEO} branch, we use an HNLF with a larger dispersion slope ($0.030 \text{ ps/nm}^2\text{-km}$) and a shorter zero dispersion wavelength (1447 nm).

The inset in Fig. 4 shows an example of the detected beat signals, which was obtained between a Ti:sapphire laser at 813 nm (Sr lattice laser) and the nearest comb modes of the 813-nm branch. The S/N was 35 dB at a resolution bandwidth of 300 kHz . For the other branches but for 798 nm , we also observed a sufficient S/N for frequency stabilization or frequency measurement.

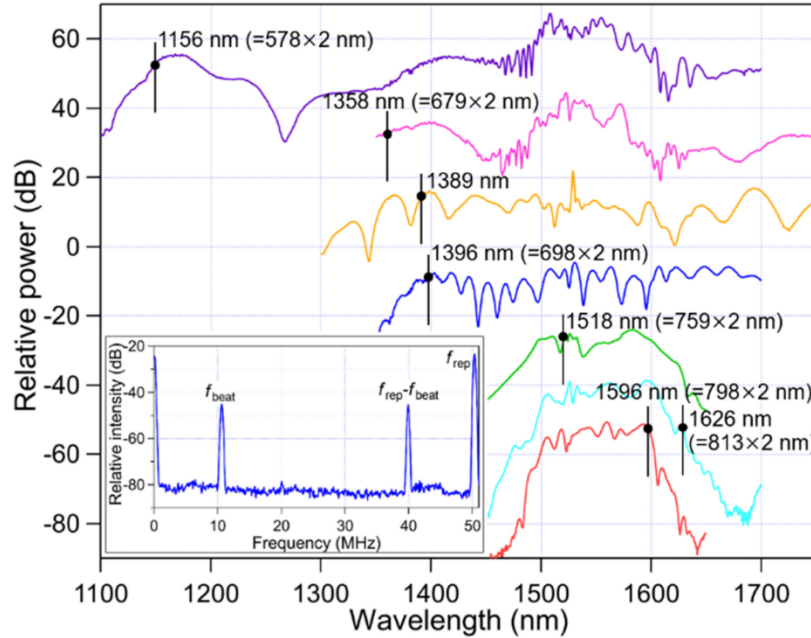


Fig. 4. Observed frequency comb spectra of the output from seven branches for different applications. The spectra are offset from each other for clarity. Each color corresponds to one branch. The black lines and dots indicate the wavelengths of the CW lasers or the fundamental lights for Sr and Yb optical lattice clocks. The inset shows the RF spectrum of the beat signal between the 813-nm Ti:sapphire laser and the comb modes (f_{beat}) observed with a spectrum analyzer. The resolution bandwidth was 300 kHz .

3.3. Frequency-locking CW lasers to 8-branch fiber comb

The detected beat signals are used to perform laser frequency offset locking with the electronic delay-line method [31,32]. The coaxial delay-line cable was 25 m long. Figure 5 shows the variation in the beat frequency between the comb and the Ti:sapphire laser (f_{beat}), which was kept frequency stabilized to the comb for about 3 hours. The counter gate time was 1 s . The frequency variation of the f_{beat} was suppressed to within 4 kHz . The inset in Fig. 5 shows the corresponding Allan standard deviation calculated from the measured variation in f_{beat} as a function of the averaging time. The stability of the beat between the frequency-locked Ti:sapphire laser and the comb was 2.2×10^{-13} for an averaging time τ of 1 s . It decreased towards 4.4×10^{-14} before 20 s while it increased towards 1.4×10^{-12} after 20 s . The observed frequency drift ($\tau > 20 \text{ s}$) is due to the drift of the reference voltage in the electric circuit for locking.

As shown in the inset in Fig. 5, the in-loop beat stability up to 10 s is better than that of UTC(NMIJ), which is used as the frequency reference of the comb. Therefore, the frequency stability of the frequency-locked Ti:sapphire laser follows UTC(NMIJ) up to 10 s , while the

instability of the locking limits the frequency stability of the offset-locked laser for a longer averaging time ($\tau > 10$ s).

The daily fluctuation of the beat in the delay-line method is investigated in [32], where a fluctuation of less than 10 kHz was demonstrated over 24 days.

For Sr and Yb optical lattice clocks operated with the lattice intensities of a few hundreds of the lattice recoil energies, a 1 MHz frequency variation in the lattice lasers corresponds to a variation of approximately 1×10^{-17} in the E1 scalar light shift of the clock transition [34,35]. The demonstrated instability of the frequency-locking CW laser shows the feasibility of the delay-line method. Other branches are also used to frequency stabilize the CW lasers used in the Sr and Yb optical lattice clocks.

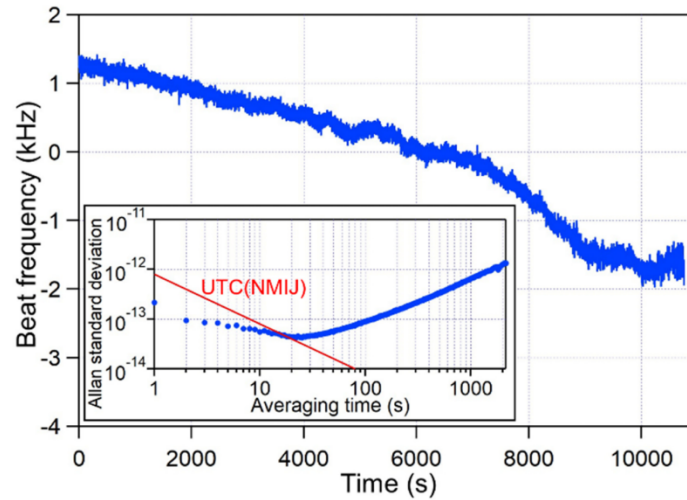


Fig. 5. Variations in the measured beat frequency (f_{beat}) between the laser and the nearest comb mode when the laser was frequency locked. The inset shows the Allan standard deviations calculated from the measured f_{beat} . The red line shows a typical frequency stability for UTC(NMIJ).

3.4. Frequency measurements of clock transition of ^{87}Sr and ^{171}Yb

We performed an absolute frequency measurement of Sr and Yb optical lattice clocks at NMIJ to check the overall reliability of the developed 8-branch frequency comb. The optical lattice clocks are detailed in [35,36].

The insets in Figs. 6(a) and 6(b) show the temporal variations of the measured frequencies of the Sr and Yb clock transitions, respectively. The corresponding Allan standard deviations are also shown in Figs. 6(a) and 6(b). The Allan standard deviation of the frequency measurement of Sr (Yb) was 9.0×10^{-14} (1.1×10^{-13}) for an 8-s averaging time, and it improved after an averaging time of 1000 s towards 2×10^{-15} (2×10^{-15}), which basically follows that of UTC(NMIJ). This result indicates that the f_{rep} of the developed 8-branch frequency comb is successfully locked to UTC(NMIJ) without frequency degradation. We note here that the frequency stabilities of the optical lattice clocks themselves are much better than that of UTC(NMIJ).

Figure 7 shows the results of several absolute frequency measurements of the clock transitions. Each measurement employed data obtained over 3-4 hours. During the measurements no comb adjustment was necessary and all the branches worked continuously as regards frequency locking and measurement. We conservatively estimated the uncertainties of the measured frequencies to be 1×10^{-14} , resulting from the dead time of the comparison between optical lattice clocks and UTC(NMIJ) [37]. The uncertainties of the optical lattice clocks are neglected here since they are well below 10^{-15} [35,36]. As shown in Fig. 7, the

results agree with the recommended frequencies by CIPM (International Committee of Weights and Measures)) [15] within our measurement uncertainties.

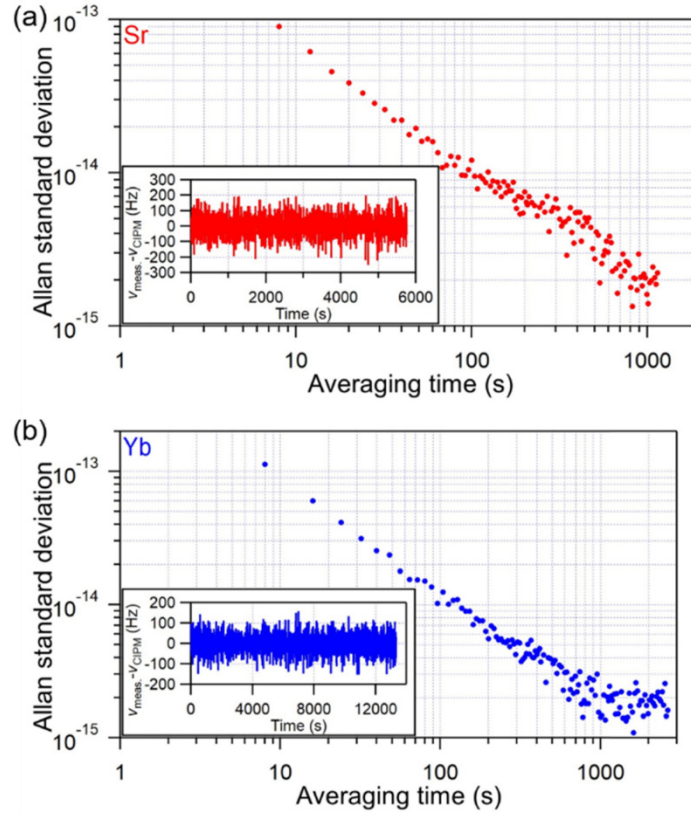


Fig. 6. Allan standard deviation calculated from the measured frequencies of (a) the 698-nm Sr and (b) the 578-nm Yb clock transitions using the comb referenced to UTC(NMIJ). The inset shows the measured frequency with an averaging time of 8 s.

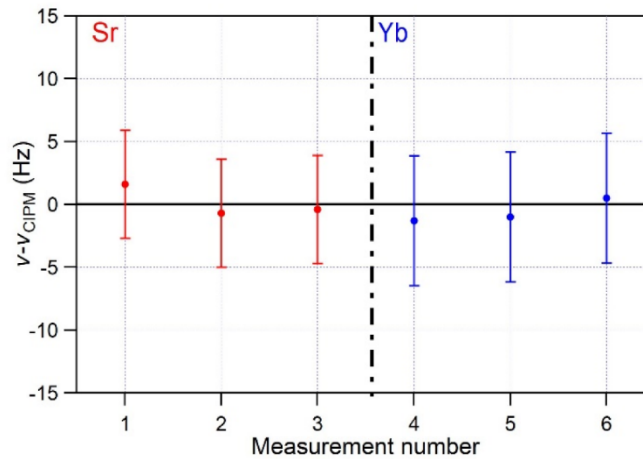


Fig. 7. Absolute frequency measurements of the 1S_0 - 3P_0 clock transition in ^{87}Sr and ^{171}Yb relative to the CIPM recommended values.

4. Discussion and conclusion

As a next step, we wanted to know how we could increase the fiber coupler stage number to $n > 3$ and obtain more application ports. An easy approach involves introducing an additional EDFA after the laser oscillator and before the fiber stages. In this way, we can have a much larger seed power before the EDFA in each branch. However, it is necessary to check the phase noise in the branches because the EDFAs in series may introduce extra phase noise that exceeds the level allowed by the applications, such as a frequency measurement or phase locking. Since frequency-locking using a delay line can be performed and used with relatively large phase noise or linewidth, it would also be a good idea to insert the series of EDFAs in one branch after the 1st fiber stage. Then we can add more stages of fiber couplers after the series of the EDFAs and obtain more application ports.

The accessible wavelengths of an Er: fiber comb usually range from 500 to 2000 nm when we use the original comb light after the HNLF or a bulk PPLN to double the frequency of the original comb light. For applications that use a light source outside the above wavelength range, for example spectroscopy in the mid-infrared [38], it would be useful to test a waveguide PPLN after the HNLF, which can provide a comb spanning more than 3.6 octaves from 0.35 to 4.4 μm [39]. Although the intensity of the spectra may not be very high due to the spread of the light energy in a large frequency span, frequency locking can be performed with a relatively small S/N.

In the present experiment, a hydrogen maser was used as the time base for the fiber comb. In some applications that only employ frequency locking, a GPS disciplined time base with a frequency stability better than 10^{-12} should be sufficient. The GPS disciplined time base allows the easier introduction of this technology into university laboratories and may also contribute to the realization of a transportable system. Towards the realization of a compact and transportable frequency stabilization system, the Er: fiber mode-locked laser using saturable absorber [40] should be a good candidate for the oscillator of a multi-branch comb system. Such a compact system may have a relatively large phase noise. But again, phase noise does not affect frequency locking using a delay line.

In some applications, we need to use narrow-linewidth combs. For example, linewidth transfer [41,42] between clock lasers and the frequency ratio measurement are important for the research of optical clocks. In such cases, a single-branch [43], a dual-branch [44] and fiber-noise-cancelling [45] Er: fiber frequency combs have advantages for the realization of low phase noise and narrow-linewidth combs.

In conclusion, we have developed an 8-branch frequency comb for laser frequency stabilization and measurement. Two repumping lasers and two lattice lasers of Sr and Yb optical lattice clocks were successfully frequency locked to the developed comb. Absolute frequency measurements of the clocks were also demonstrated by using two of the branches in the comb. We have confirmed that the multi-branch frequency comb worked continuously during the long-term operation of the clocks. The developed multi-branch frequency comb has a low cost and is reliable for use in many applications including optical lattice clocks and quantum optics experiments.

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